SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: **Imitation (Level 1)** **Total Marks: 25**

Question Count: 5

Code of Conduct

I G. Piyusha (192472377) (Name / Reg. No.)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Piyusha

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. *Trace the evolution of cloud computing from hardware evolution to internet software evolution,* and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

Define cloud computing & explain any four characteristics that make it different from traditional on-premise computing

Cloud computing is the delivery of computing resources such as servers, storages, databases, networking, software & processing Power over the internet, usually on a pay-as-you-use bias

Four characteristics

1. On-demand self-service:

Users can provision computing resource such as storage or virtual machines whenever required without direct interaction with the service provider.

Ex: Creating a virtual machine in AWS within minutes.

2. Broad network access:

Cloud services are accessible through the internet using devices such as laptops, smartphones & tablets.

Ex: Accessing Google Drive from a phone or laptop.

3. Resource pooling:

Cloud providers share a large pool of computing resources among multiple customers using virtualization while keeping their data logically isolated

Ex: Multiple organizations using virtual machines on same physical server.

4. Rapid elasticity:

Resources can be quickly increased or decreased accordingly to demand.

Ex: An online shopping website can increase server capacity during a festival sale & reduce it afterward.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution & show how these changes enabled modern cloud platforms.

1. Hardware evolution:

Early computers were large & expensive mainframes used by multiple users through terminals. Later, improvements in processors.

2. Distributed computing:

Computing tasks began to be distributed across multiple connected computers instead of depending on a single machine.

3. Internet development:

The growth of the internet & high-speed networks made it possible to access computing resources & applications from anywhere.

4. Virtualization:

Virtualization allowed one physical server to run multiple virtual machines. The improved hardware utilization & made computing resources easier to allocate dynamically.

5. Web & Software evolution:

Users could access software through browsers without installing it locally.

Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology & cloud computing as a service model.

Server virtualization divides one physical server into multiple virtual machines using a hypervisor.

Role: It improves resource utilization, provides isolation, & allows flexible resources allocation.

Difference:

- Virtualization: Technology that creates virtual resource
- Cloud computing: Service model that delivers computing resources over the internet.

Ex: VirtualBox creates virtual machines : AWS provides cloud services.

4. Discuss the role of data centres in delivering cloud services

A data center contains servers, storage, networking, power, & security systems.

Roles:

- Stores & processes data
- Runs cloud applications
- Provides Internet connectivity
- Ensure availability & backup
- Supports scalability.

Ex: Google data centers support google drive & other cloud services.

5. Differentiate Iaas, Paas, Saas & Xaas with one suitable real-world example for each model.

1. Iaas (Infrastructure as a Service):

Iaas provides virtualized computing resources such as servers, storage & networking over the internet. Users can manage the operating system & applications.

2. Paas (Platform as a Service):

Paas provides a platform with tools & an environment for developers to build, test, & deploy applications without managing the underlying infrastructure.

3. Saas (Software as a Service):

Saas provides ready-to-use software over the internet. Users can access the software through a web browser.

4. Xaas (Anything as a Service):

Xaas is a board term that refers to various IT services delivered over the internet on demand, usually through a subscription or pay-as-you-go model.

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